

# Public transportation trip flow modeling with generalized regression neural networks

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## Abstract

Artificial neural networks (ANNs) are one of the recently explored advanced technologies, which show promise in the area of transportation engineering. The presented study comprised the employment of this seldom used ANN method, generalized regression neural network (GRNN), in comparison to both a frequently applied neural network training algorithm, feed-forward back-propagation (FFBP), and a stochastic model of auto-regressive structure for the purpose of forecasting daily trip flows, which is an essential component in demand analysis. The study is carried out under the motivation of knowing that modeling daily trips for available transportation modes will facilitate the arrangement for effective public infrastructure investments and the cited papers in the literature did not make use of and handle any comparison with GRNN method. The ANN predictions are found to be quite close to the observations as reflected in the selected performance criteria. The selected stochastic model performance is quite poor compared with ANN results. It is seen that the GRNN did not provide negative forecasts in contrast to FFBP applications. Besides, the local minima problem faced by FFBP algorithm is not encountered in GRNNs.

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## 1. Introduction

The urbanization concept is mainly defined by the attributes specifying the quality and the characteristics of life standards that are available for the inhabitants of an urban area. Complexly structured urban transport systems, the crucial attribute for sustainable development on a settlement, require the determination of the level of provided infrastructure. The way managing the infrastructure affects the demand for travel, which can vary by transportation mode, location, and time of day, in turn affects the performance of the transportation system. In the longer term, transport system performance affects land use patterns

and this further influences demand. In the short term, both the system performance and the demand patterns produce both benefits and inconveniences for users, residents and businesses. Therefore, it is important to ensure that any infrastructure investment will have beneficial effects on the overall transport system and those affected by it [1]. Because of the infrastructure and the superstructure concept of transportation being generally irreversible, the demand analysis for an investment in a system approach is crucial. By defining the demand for transportation as a potential for traffic flow, the importance of the trips made within a day is being clarified.

Studies considering the daily trip flows in the past, be it time series models [2,3] and Kalman filtering models [4], were employed to forecast traffic volume. Due to the stochastic nature of traffic flow and the strongly non-linear characteristics of traffic dynamics, methods of soft computing have received much attention since early 90s and

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considered as alternatives for the traditional statistical models. Among these methods the neural network (NN) approaches have been commonly applied in a number of areas of transport [5]. Including the studies of traffic volume forecasting [6,7], short-term traffic flow prediction [8–11], macroscopic modeling of freeway traffic [12], hybrid models of NNs [13,14], and NN applications in mode choice modeling (summarized in [15]), diverse kinds of NNs have been proposed in the literature.

The artificial neural network (ANN) approach, which is a non-linear black box model, seems to be a useful alternative for modeling the complex time series. In the majority of the above-mentioned cited papers, feed-forward back-propagation method (FFBP) was employed to train the neural networks. The performance of feed-forward back-propagation neural networks (FFBPNNs) was found superior to conventional statistical and stochastic methods in continuous flow series forecasting. However the FFBP algorithm has some drawbacks. It is very sensitive to the selected initial weight values and may provide performances differing from each other significantly. Another problem faced during the application of FFBP algorithm is the local minima issue. During the training stage the networks are sometimes trapped by the local error minima preventing them to reach the global minimum. Methods used in the literature to overcome local minima problem as training a number of networks starting with different initial weights, the on-line training mode to help the network to escape local minima, inclusion of the addition of random noise, employment of second order (Newton algorithm, Levenberg–Marquardt algorithm), global optimization method or other global methods (stochastic gradient algorithms, simulated annealing) are summarized in the literature [16]. In the review study of the ASCE Task Committee other ANN methods such as conjugate gradient algorithms, radial basis function, cascade correlation algorithm and recurrent neural networks are briefly explained [17].

For knowing that modeling daily person trips for available transportation modes will facilitate the arrangement for effective public infrastructure investments and the above-mentioned papers did not make use of and handle any comparison with GRNN method, we model daily person trips made by public transit mode by means of two ANN methods, FFBPNN and generalized regression neural network (GRNN), used as forecasting tools in comparison to an auto-regressive model throughout this study.

The article consists of four sections. After the introduction, ANN concept and utilized methods of ANNs, FFBPNN and GRNN, are explained briefly in Section 2. Data analysis, network preparations, and prediction results are presented in Section 3. The paper ends with a discussion of the findings and possible future extensions in Section 4.

## 2. Neural networks and used methods

In this section, theoretical basis of the utilized neural NN methods are summarized in brief.

### 2.1. Neural network theory

In the neural network model, the neuron is the basic component. As shown in Fig. 1, a multilayer perceptron structure is a fully interconnected set of layers of neurons. Each neuron ( $N_{i..j}$ ) of a layer is connected to each neuron of the next layer so that only forward transmission through the network is possible, from the input layer to the output layer through the hidden layers.

In this structure, the output  $Y_i$  of each neuron of the  $n$ th layer is defined by a differentiable non-linear function  $F$  as shown with Eq. (1).

$$Y_i = F\left(\sum_j w_{ij}y_j\right) \quad (1)$$

where  $F$  is the non-linear activation function,  $w_{ij}$  are the weights of the connection between the neuron  $N_j$  and  $N_i$ ,  $y_j$  is the output of the neuron of the  $(n-1)$ th layer. For each input vector presented into the network, the set of connection weights between the different neurons determines the response of the network in the output layer. Partial specifications of the problem (i.e. input–output pairs) allow the measurement of output error of the network and the adjustment of its behavior. An iterative algorithm does the adjustment during training phase of the neural network. During the training phase, a selected set of training patterns is presented to the network. When the error between the network output and the desired output is minimized, the network can be used in a testing phase with test pattern vectors. At this stage, the neural network is described by the optimal weight configuration, which means that theoretically ensures the output error minimization. The average system error or the mean squared error (MSE),  $E$ , for all input patterns is shown with Eq. (2).

$$E = \frac{1}{2N} \sum_{n=1}^N \sum_{i=1}^m (d_{ni} - y_{ni})^2 \quad (2)$$

Here  $d_{ni}$  is the system output value for the  $n$ th pattern,  $y_{ni}$  is the neural network output value for the  $n$ th pattern, and  $N$  and  $m$  are the total numbers of the inputs and outputs respectively.

As the form of the mapping  $F$  a priori is not known, an approximation is sought. The development of ANNs offers

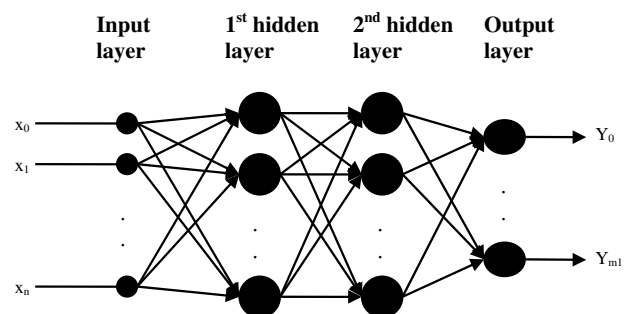


Fig. 1. Configuration of a feed-forward multilayer perceptron.

an alternative to approximate such multivariate functions and had been widely studied beginning from the late 80s with various results, concerning density or complexity, established for different situations [18–23].

## 2.2. Feed-forward back-propagation neural network (FFBPNN) method

As the form of the mapping  $F$  (Eq. (1)) a priori is not known, an approximation is sought. The development of ANNs offers an alternative to function approximators. The back-propagation NN, which has only a finite number of parameters, can approximate most bounded functions with arbitrary precision [18] and is used here to approximate  $F$ . A multilayer feed-forward back-propagation neural network  $\text{FFBPNN}(\cdot) \in R^m$  is a mapping constructed recursively by a vector linear/non-linear functions, shown with Eq. (3).

$$\text{FFBPNN}(\Psi, \mathbf{W}, \Theta) = \Psi[\mathbf{W}_n \Psi(\mathbf{W}_{n-1} \Psi(\cdots \Psi(\mathbf{W}_1 \mathbf{U} \mathbf{I} + \theta_1) + \theta_{n-1}) + \theta_n] \quad (3)$$

Here  $n$  is the number of layers;  $\mathbf{W}_i$  is the linear operator of proper dimensions where  $i = 1, \dots, n$ ;  $\theta_i$  is the bias vector, where  $i = 1, \dots, n$ ;  $\mathbf{U} \mathbf{I}$  is the input vector of the neural network;  $\Psi$  is the vector-valued linear/non-linear mapping (transfer function);  $\mathbf{W} = [\mathbf{W}_1, \dots, \mathbf{W}_n]$ ; and  $\Theta = [\theta_1^t, \dots, \theta_n^t]^t$  ( $t$  denotes the transpose of vector/matrix).

A NN is non-linear if at least some of its transfer functions are non-linear. A typical non-linear transfer function is the sigmoidal function, as given by Eq. (4).

$$\Psi_{a,b}(x) = \frac{1}{1 - e^{-ax+\beta}} \quad (4)$$

The weight matrices  $\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_n$  and bias vectors  $\theta_1, \theta_2, \dots, \theta_n$  are adjusted by a learning rule called back-propagation, to minimize output error with respect to the weights and thresholds [24].

## 2.3. Generalized regression neural network (GRNN) method

General regression neural network, Specht's term [25] for Nadaraya–Watson kernel regression [26,27], also reinvented in the NN literature by Schioler and Hartmann [28], does not require an iterative training procedure as in back-propagation method. It approximates any arbitrary function between input and output vectors, drawing the function estimate directly from the training data. Furthermore, it is consistent; that is, as the training set size becomes large, the estimation error approaches to zero, with only mild restrictions on the function. The GRNN is used for estimation of continuous variables, as in standard regression techniques. It is related to the radial basis function (RBF) networks, networks that have the universal approximation property [29], arise naturally as regularized solutions of ill-posed problems [30] and are dealt well in the theory of interpolation [31], and is based on a standard sta-

tistical technique called kernel regression. RBF networks' simple structure enables learning in stages, gives a reduction in the training time, and this has lead to the application of such networks to many practical problems. The adjustable parameters of such networks are the centre (the location of basis functions), the width of the receptive fields (the spread), the shape of the receptive field and the linear output weights. GRNN can be treated as a normalized RBF network in which there is a hidden unit centred at every training case. These RBF units are called "kernels" and are usually probability density functions such as the Gaussian. By definition, the regression of a dependent variable  $y$  on an independent  $\mathbf{x}$  estimates the most probable value for  $y$ , given  $\mathbf{x}$  and a training set. The regression method will produce the estimated value of  $y$  that minimizes the MSE. GRNN is a method for estimating the joint probability density function of  $\mathbf{x}$  and  $y$ , given only a training set. Because the probability density function is derived from the data with no preconceptions about its form, the system is perfectly general.

By following Specht [25],  $f(\mathbf{x}, y)$  represents the known joint continuous probability density function of a vector random variable,  $\mathbf{x}$ , and a scalar random variable,  $y$ , the conditional mean of  $y$  given  $\mathbf{X}$ , the regression of  $y$  on  $\mathbf{X}$ , is given by Eq. (5).

$$E[y|\mathbf{X}] = \frac{\int_{-\infty}^{\infty} y f(\mathbf{X}, y) dy}{\int_{-\infty}^{\infty} f(\mathbf{X}, y) dy} \quad (5)$$

when the density  $f(\mathbf{x}, y)$  is not known, it must usually be estimated from a sample of observations of  $\mathbf{x}$  and  $y$ . The probability estimator  $f'(\mathbf{X}, Y)$  (shown in Eq. (6)) is based upon sample values  $\mathbf{X}^i$  and  $Y^i$  of the random variables  $\mathbf{x}$  and  $y$ , where  $n$  is the number of sample observations and  $p$  is the dimension of the vector variable  $\mathbf{x}$ .

$$\begin{aligned} f'(\mathbf{X}, Y) &= \frac{1}{(2\pi)^{(p+1)/2} \sigma^{(p+1)}} \\ &\times \frac{1}{n} \sum_{i=1}^n \exp \left[ -\frac{(\mathbf{X} - \mathbf{X}^i)^T (\mathbf{X} - \mathbf{X}^i)}{2\sigma^2} \right] \\ &\times \exp \left[ -\frac{(Y - Y^i)^2}{2\sigma} \right] \end{aligned} \quad (6)$$

A physical interpretation of the probability estimate  $f'(\mathbf{X}, Y)$  is that it assigns sample probability of width  $\sigma$  for each sample  $\mathbf{X}^i$  and  $Y^i$ , and the probability estimate is the sum of those sample probabilities [25]. Defining the scalar function  $D_i^2$  (shown with Eq. (7)) and performing the indicated integrations yields Eq. (8).

$$D_i^2 = (\mathbf{X} - \mathbf{X}^i)^T (\mathbf{X} - \mathbf{X}^i) \quad (7)$$

$$Y'(\mathbf{X}) = \frac{\sum_{i=1}^n Y^i \exp \left( -\frac{D_i^2}{2\sigma^2} \right)}{\sum_{i=1}^n \exp \left( -\frac{D_i^2}{2\sigma^2} \right)} \quad (8)$$

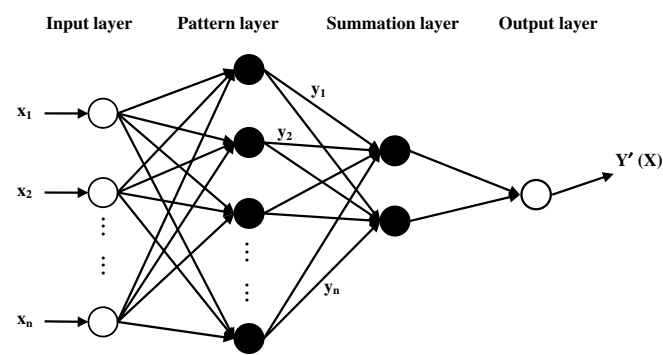


Fig. 2. Schematic diagram of a generalized regression neural network (GRNN) architecture.

The resulting regression (Eq. (8)) is directly applicable to problems involving numerical data. When the smoothing parameter  $\sigma$  is made large, the estimated density is forced to be smooth and in the limit becomes a multivariate Gaussian with covariance  $\sigma^2\mathbf{I}$ . On the other hand, a smaller value of  $\sigma$  allows the estimated density to assume non-Gaussian shapes, but with the hazard that wild points may have too great an effect on the estimate [25]. The schematic diagram of a GRNN architecture is shown in Fig. 2.

### 3. Analysis process

#### 3.1. Information on data set

Since a large number of traffic data are observed in a time series structure, a collection of sequential observations, time series analysis has been a useful tool for traffic flow prediction. In this study, public transport (PT) trip flow the averages per half an hour per day using AKBIL (an electronic ticketing system integrated to all PT modes) in Istanbul metropolitan area (Turkey) are used as time series data. Data of PT trip flows vary within public bus, tram, light rail transit, metro, sea-bus, and city marine line modes and are obtained from the observations of the first 11 months of the year 2002 starting at the January 1st and ending at the November 30th. The trip flows are considered as person-trips made by commuters. Data storage had been done continuously for 24 h in a day by an electronic device that is used to check tickets at vehicle entrance. Since collected data is continuous, an aggregation level of half an hour is selected, hence, 48 data are processed for one day within the above-mentioned data collection period. Table 1 summarizes the descriptive statis-

Table 1  
Information on analyzed data set

| Type of the data                       | Region                     | UI <sub>min</sub> | UI <sub>max</sub> | Mean   | Standard deviation | Skewness | Observation period    |
|----------------------------------------|----------------------------|-------------------|-------------------|--------|--------------------|----------|-----------------------|
| Trip flow (lag: 1/2 hour; unit: trips) | Istanbul Metropolitan area | 15                | 2,367,412         | 521.13 | 451.78             | 0.65     | 01.01.2002–30.11.2002 |

UI<sub>min</sub> minimum observation value, UI<sub>max</sub> maximum observation value.

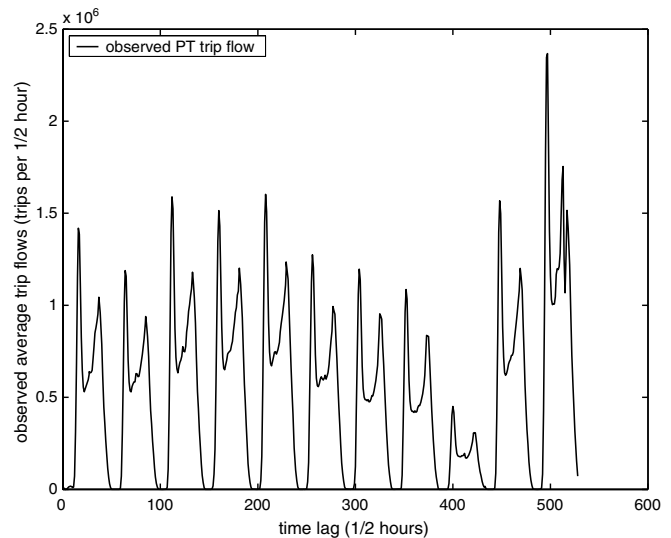


Fig. 3. Time series plot of average public transport trip flows per half an hour.

tics of the analysed data set and Fig. 3 shows the plot of the time series data.

The last 348 values of average PT trip flow per half an hour (the interval starts at the April 1st and ends at the November 30th in the year 2002) are analyzed during the training stage of both two neural networks. The first 180 values (the interval starts at the January 1st and ends at the March 30th in the year 2002) are then used for the testing phase. The reason for considering the last part of the time series as training data is the higher and lower period maximum and minimum values, respectively, compared with testing data.

#### 3.2. ANN preparations and prediction results

As the objective of the study is the forecast of the trip flow averages per half an hour by using ANNs, the analysis process consisted of two steps, which are training and testing respectively. In order to simulate these steps, the prediction problem is transformed into the following minimum norm problem: The search for FFBPNN and GRNN configurations that will minimize the Euclidean distance,  $\|P(\mathbf{UI}_k) - \text{NN}(\mathbf{UI}_k)\|$ , where  $\mathbf{UI}_k$  and  $\mathbf{Y}_k$  are the observation vector series,  $\mathbf{UI} = [\mathbf{UI}_{t-1}, \mathbf{UI}_{t-2}, \mathbf{UI}_{t-3}, \mathbf{UI}_{t-4}]$  is the input vector of NN, and  $\mathbf{Y}_{k+1} = P(\mathbf{UI}_k)$ . Codes for GRNN and FFBPNN simulations are written in C language and run on a computer with the following features: processor,



AMD Turion64 MT28 (1.6 GHz); memory, 448 MB (533 MHz DDR SDRAM); hard drive, 60 GB (4200 rpm). Throughout the NN simulations, finding the optimal configurations of NNs was the aim of the training period, and then the selected networks were applied to the testing data.

In the light of the NN methods, the solution to the minimum norm problem involves a number of steps. The first one is the choice of the model inputs, and the second step is the attainment of parameters that minimize the norm given above. The number of input nodes to the NN model is dependent on the number of the time lags that is determined by the analysis of the autocorrelation function [32] belonging to examined time series data set. The correlation between two random variables measures their linear association. For a random sequence indexed by time, the correlation becomes a function of time. The autocorrelation function (ACF)  $\rho_{UIUI}(\tau)$  of a random process, say  $UI_k$  for our case, of lag  $\tau$  can be obtained by Eq. (9).

$$\rho_{UIUI}(\tau) = \frac{E[(UI_{k-\tau} - E[UI_{k-\tau}])(UI_k - E[UI_k])]}{\sqrt{E[(UI_{k-\tau} - E[UI_{k-\tau}])^2]E[(UI_k - E[UI_k])^2]}} \quad (9)$$

where  $E$  is the expectation operator [33]. Reckoning on to Schwartz inequality, it can be said that  $UI_k$  and  $UI_{k-\tau}$  are converging to be perfectly correlated if  $\rho_{UIUI}(\tau) \cong \pm 1$ . Four time lags were found sufficient; hence, the input layer consisted of four nodes representing consecutive trip flow averages per half an hour,  $UI_{t-1}$ ,  $UI_{t-2}$ ,  $UI_{t-3}$ ,  $UI_{t-4}$ . The plot of the ACF belonging to analyzed data set is shown in Fig. 4.

In order to capture a better performance and more optimal convergence, the values of the training inputs and outputs are scaled between zero and 1 as in Eq. (10), where  $UI_i^{\text{scaled}}$ ,  $UI_i^{\text{actual}}$ ,  $UI_{\max}$  and  $UI_{\min}$  are the scaled value of the observed variable, actual value of the observed vari-

able, maximum observation value of the data set and the minimum observation value of the data set respectively.

$$UI_i^{\text{scaled}} = \frac{UI_i^{\text{actual}} - UI_{\min}}{UI_{\max} - UI_{\min}} \quad (10)$$

For the iterative structure of the training process needs a threshold value to stop learning, performance criteria for varying NN configurations require convergence to some selected error term targets. MSE values are targeted for GRNN and FFBPNN training processes.

### 3.2.1. Preparation of FFBPNN configuration

Initially FFBPNN method is used for trip flow prediction. The Levenberg–Marquardt algorithm [34,35] is utilized to minimize output error with respect to the weights and thresholds in FFBPNN simulations. The determination of the number of the hidden layers providing best training results was the initial process of the training procedure, following the selection of model inputs explained in Section 4. Selection of layers and transfer function  $\Psi$ , and the final step of parameter attaining to minimize the norm were then performed. The MSE and the ACF are used as criteria to evaluate the performance of the training simulations.

Because the last two steps are largely trial-and-error processes and experiments involving neural networks with input nodes more than 4 and hidden units more than 3 did not show any sizeable improvement in prediction accuracy, a three-layer, three hidden units back-propagation neural network with the sigmoidal function  $\Psi_{\alpha,\beta}(x)$  as it is non-linear transfer function is selected (FFBP(4,3,1)). The unique output layer node represented the averages of trip flows per half an hour at time  $t$ ,  $U_t$ , to be predicted.

In the training process, the iteration number 90 is found sufficient in respect to a fixed MSE value. After the completion of the training stage, testing stage took place. The optimum learning,  $\eta$ , and the momentum,  $\alpha$ , rates are determined after trials observing the MSE produced at the end of the training stage. It is seen that  $\eta$  and  $\alpha$  should be decreased if the number of the input and output layers are increased. On the other hand, the iteration number increases by decreasing  $\eta$  and  $\alpha$  values.

### 3.2.2. Preparation of GRNN configuration

In a GRNN structure the hidden weights to output weights are the target values, so the output is simply a weighted average of the target values of training cases close to the given input case. The only weights need to be learned are the widths of the RBF units. Optimization of these widths, the smoothing parameters, and the minimization of the MSE on GRNN generated values by means of the estimated value of  $UI_k$  produced by regression method are the aim of the training process. The smoothing parameter (the width of the RBFs') is selected 0.51 regarding to the minima existed corresponding to calculated MSE value.

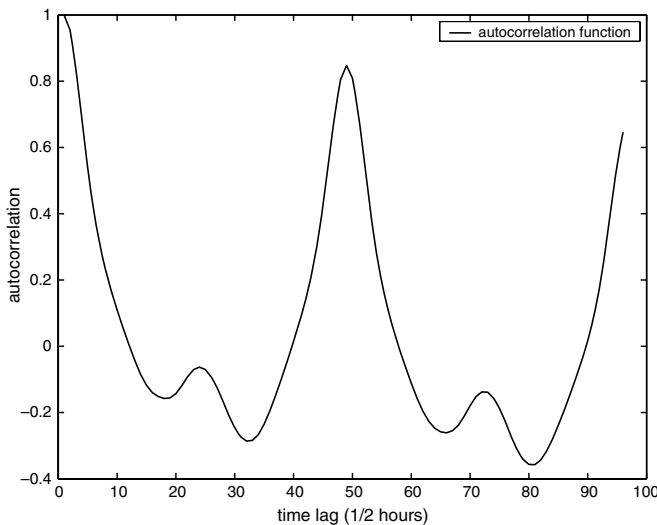


Fig. 4. Plot of autocorrelation function belonging to analyzed data set.

Following the training period the network is applied to the testing data. The regression had after the integrations in Eq. (5) and substitutions of the scalar function and smoothing parameter is directly applied to testing data.

### 3.3. ANN prediction results and compared stochastic model

The MSE values and the determination coefficients for different orders of time series depending on the optimum

Table 2  
MSE and coefficient of determination values of FFBPNN and GRNN predictions (testing period)

| Input layer<br>node number | FFBPNN   |       | GRNN     |       |
|----------------------------|----------|-------|----------|-------|
|                            | MSE      | $R^2$ | MSE      | $R^2$ |
| 1                          | 0.003021 | 0.905 | 0.002032 | 0.916 |
| 2                          | 0.000490 | 0.985 | 0.000844 | 0.988 |
| 3                          | 0.000440 | 0.987 | 0.000403 | 0.990 |
| 4                          | 0.000333 | 0.990 | 0.000316 | 0.995 |
| 5                          | 0.000421 | 0.980 | 0.000367 | 0.982 |

$R^2$  determination coefficient.

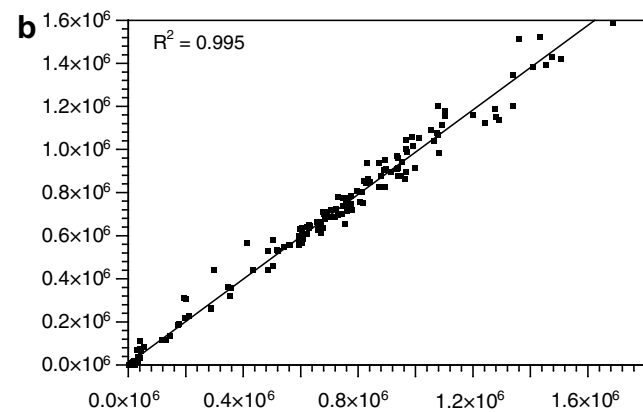
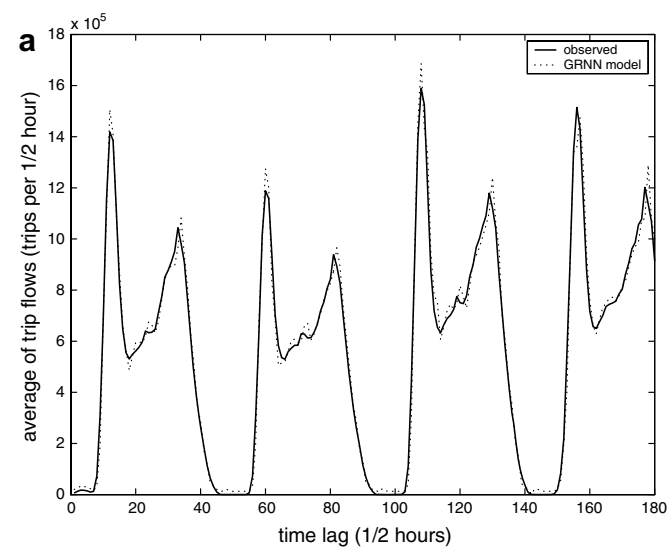


Fig. 5. Comparison of observed PT trip flow averages per half an hour with GRNN forecasts in testing period plotted as time series (a), and scatter plot (b).

hidden unit number for each order is shown in Table 2. The MSE values for the testing period of each neural network are found as 0.000316 and 0.000412 respectively for the GRNN and the FFBPNN. It is clear that the best performance criteria are obtained for a GRNN configuration with four input nodes (Table 2). The testing period predictions provided by GRNN and FFBPNN are compared with observations in Figs. 5 and 6, respectively. Note that the scatter plot graphs given in this study are plotted with the forecasted values and the corresponding observed values. The time series and scatter plots showed a quite satisfactory agreement between ANN and observation values.

Following the ANN forecasts a comparison is made by means of a stochastic model. In the statistical field, the auto-regressive (AR) models are said to be optimal in the sense that no other uni-variate forecasting tools provide smaller values on terms of error. However, this is valid only for the uni-variate models that are linear combinations of the past values belonging to time series with fixed coefficients. Furthermore, the Box–Jenkins approach relies heavily on training data correlation structure. But both the

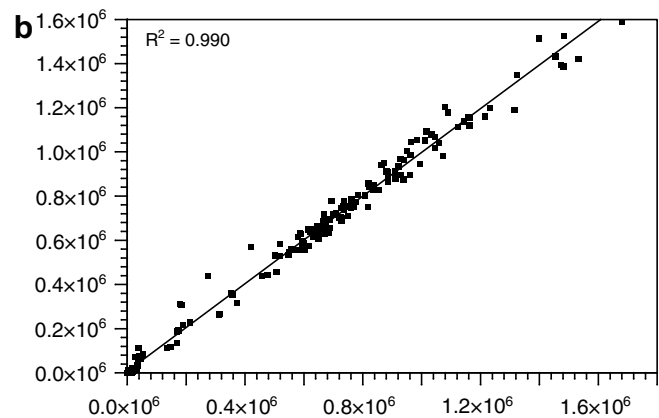
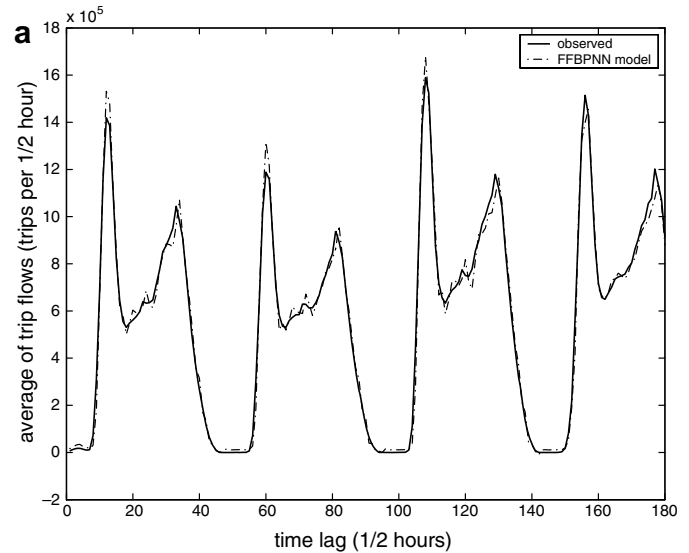


Fig. 6. Comparison of observed PT trip flow averages per half an hour with FFBPNN forecasts in testing period plotted as time series (a), and scatter plot (b).

increasing computing power and the widespread use of information criteria, such as the Akaike information criterion considered in this study and the Schwarz Bayesian criterion, for model selection have removed much of the subjectivity. So, an auto-regressive model of order four is used in order to have a comparison with a stochastic model. The AR model of order  $p$ , i.e.,  $AR(p)$ , is given by Eq. (11), if there exist constants  $\delta, \varphi_1, \varphi_2, \dots, \varphi_p$  and a white noise  $\{\varepsilon\}$  such that Eq. (11) is satisfied [36]:

$$UI_t = \delta + \varphi_1 UI_{t-1} + \varphi_2 UI_{t-2} + \dots + \varphi_p UI_{t-p} + \varepsilon_t \quad (11)$$

where  $UI_t$  is the variable at time  $t$  of the sequence of zero mean and unit variance,  $UI_{t-1}, UI_{t-2}, \dots, UI_{t-p}$  are the previous values of the variable  $y_i$ ;  $\varepsilon_t$  is the random number at  $t$ th time; and  $\varphi_1, \varphi_2, \dots, \varphi_p$  are the auto-regressive parameters. Data period between the April 1st and the November 30th in the year 2002 (training period of ANNs) is used to obtain the AR(4) model. Likewise, data period between the January 1st and the March 30th in the year 2002 (testing period of ANNs) is used to make forecast with the AR(4)

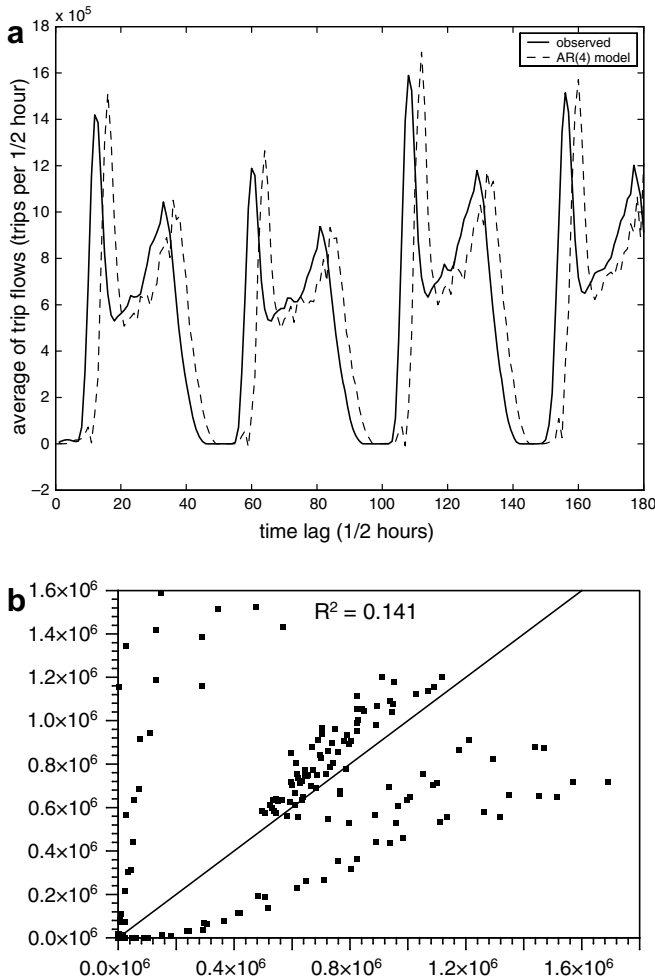


Fig. 7. Comparison of observed PT trip flow averages per half an hour with AR(4) model forecasts in testing period plotted as time series (a), and scatter plot (b).

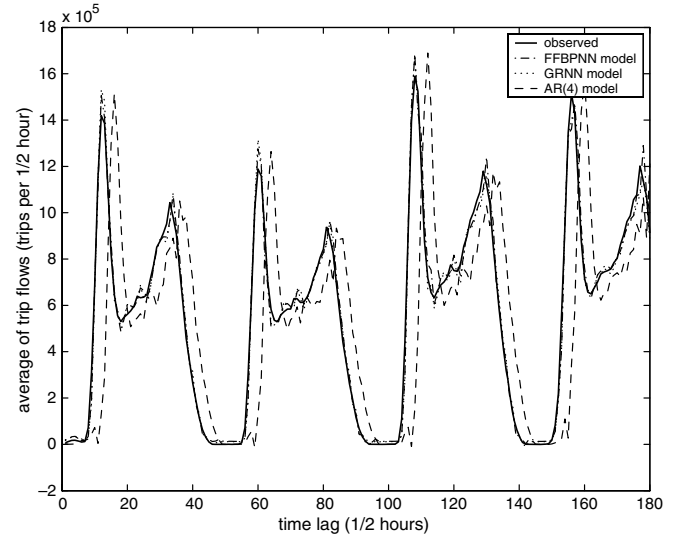


Fig. 8. Overall comparison of observed PT trip flow averages per half an hour with GRNN, FFBPNN, and AR(4) model forecasts.

model. The results are presented in Fig. 7. The AR(4) is indicated in Eq. (12).

$$UI_t = 2.05121UI_{t-1} - 1.52353UI_{t-2} + 0.41155UI_{t-3} + 0.04237UI_{t-4} \quad (12)$$

The MSE obtained from the stochastic model is equal to 0.03025 which is significantly higher compared with the corresponding ANN values (0.000316 and 0.000412). The scatter plot illustrates clearly the deviations from observations (Fig. 7b). The results point out that the ANNs forecasts for the observed values are closer to the original ones compared with the AR(4) model. To present the difference between three forecasting methods (FFBP, RBF and AR(4)) clearly, the testing period forecasts were plotted with the corresponding observations in Fig. 8.

FFBPNN simulations were required until obtaining a satisfactory performance criterion value since FFBPNN method performance is very sensitive to the randomly assigned initial weight values. Besides requiring quite a long duration for successful simulations, learning with Levenberg–Marquardt algorithm, an efficient gradient descent training algorithm, caused local minima problem in FFBPNN trials, plots of which are not given in this paper due to space limitation. Also note that the selected FFBPNN configuration generated negative forecasts regarding to Figs. 6a and 8.

#### 4. Conclusions

The superiority of GRNNs over a conventional method in the reviewed prediction study can be attributed to the capability of ANNs to capture the non-linear dynamics and generalize the structure of the whole data set. ANNs' simulations are easy to code and implement for constructing intricate multipurpose non-linear solutions. The

method has no limitations in the form of fixed assumptions or formal constraints. The NN has a distributed processing structure. Each individual processing unit or the weighted connection between two units is responsible for one small part of the input–output mapping system. Therefore, each component has no more than a marginal influence with respect to the complete solution and that the mechanism will still function and generate reasonable mappings.

Modeling public transport trip flows and having accurate predictions help to determine the future demand for modal choice which leads to the focus of forthcoming investments for an urban area. In this study a GRNN trip flow estimating model is developed and compared with both an FFBPNN and a stochastic model, AR(4), by computing the field data for Istanbul metropolitan area. Among the studied models, significant improvements in the prediction are made by the NN models due to NNs' flexibility to adapt to time series data. The provided estimates of the continuous variables and convergence to underlying regression surface fit the concept of time series. GRNNs performed faster learning by not requiring an iterative procedure in contrast to FFBPNN method. Besides, the proposed GRNN did not face the local minima problem by providing prediction results with a single iteration, where FFBPNN configuration failed by being trapped to local minima during simulation trials. Making forecasts by generating negative values after trials in FFBP algorithm trained NNs is not encountered in GRNN simulations. The GRNNs succeeded by generating non-negative values and closer estimates.

The model configuration proposed in this study can be adopted as an integral part of an advanced traffic management system (ATMS), the goal of which is to predict traffic demand as accurately as possible, as a real world application. This makes an ATMS practise result in providing better measurements (e.g., increase the frequency of public transit mode during peak hours) based on such predictions.

Possible future extensions of the study will be on the application of the hybrid models of neural networks.

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